```
int f(int x, int & y);
int main()
{   int z = 3, w = 27;
    int r = f(w, z);
    :
}

int f(int x, int & y)
{   :
}
```

Terminology

Prototype

Formal parameters

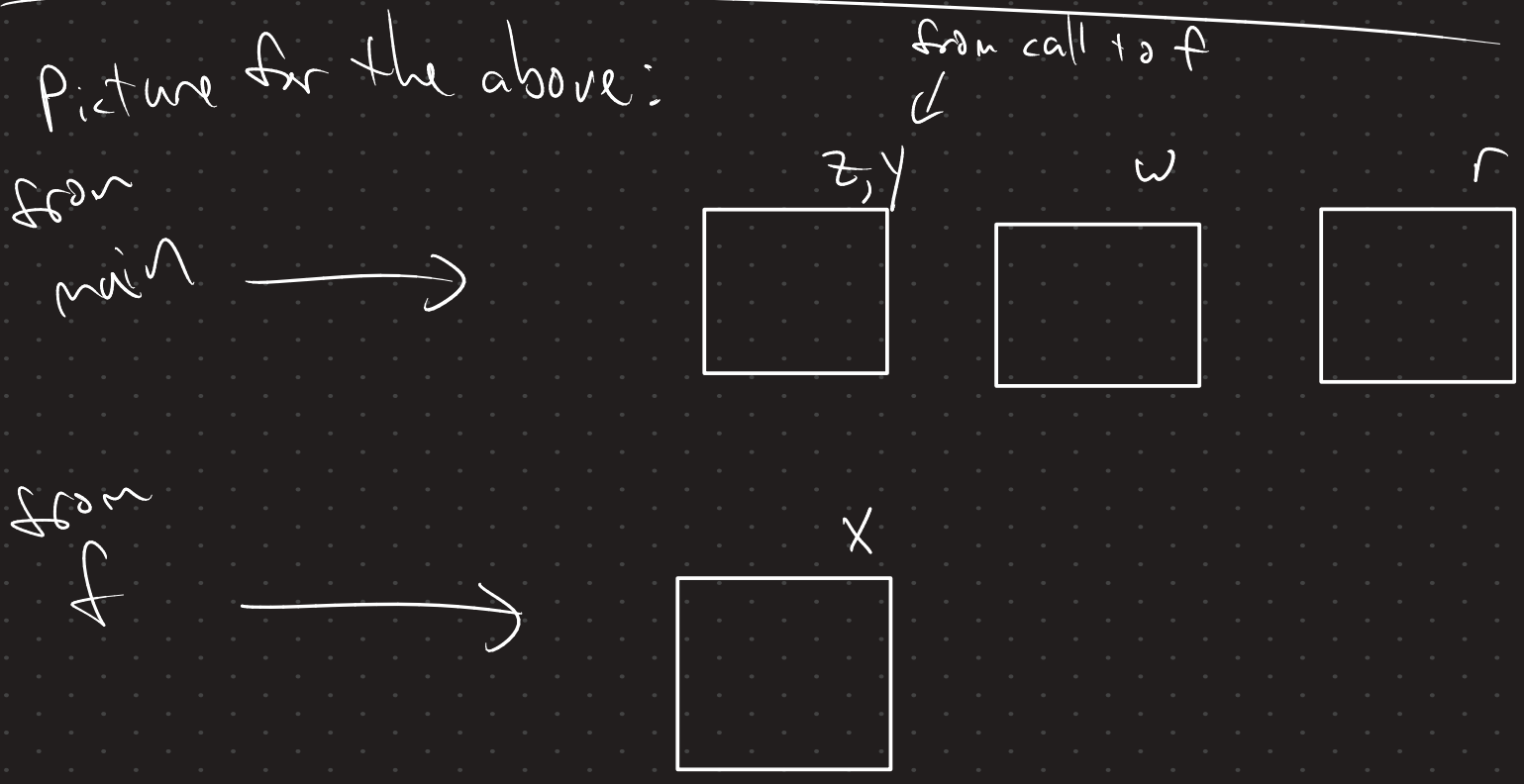
Actual parameters

Return type

Function header

Function call

Picture for the above:



Q: how to return multiple values?

E.g., how to model

$$f : \mathbb{Z} \longrightarrow \underline{\mathbb{Z} \times \mathbb{R}} \quad ?$$

A: There are many options!

① Make your own datatype, like IntDoublePair...

② Use reference parameters & have the caller supply the space where the results will be written:

Product: $\{(z, r) \mid z \in \mathbb{Z}, r \in \mathbb{R}\}$

$(2, 7.1), (3, 9.9), (2, 1.0) \ldots$

```
void f(int x, int & y, double & z);
```

input

output

Example:

```
int main()
{   int Y;
    double Z;   // f will write its answer here!
    f(23, Y, Z);
```

// Now Y, Z have been written by f.

}

Note: Instead of return (a, b),
f will do this:

```
y = a;
z = b;
return;
```

New Topic Vectors

Problem: read #s from stdin and print them in reverse order.

~~int x, y, z, w, a, b;~~

Issue: we don't know how many variables we'll need ahead of time (compile time)

Vectors give you an expandable container of variables.